THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** drop objects of different masses from the same height	Test by a controlled experiment.
2	**(c)** pumpkin: creeper	Pumpkin has a weak stem that spreads.
3	**(d)** protein and vitamins	Polished rice mostly has starch.
4	**(a)** either a magnet or magnetic material	Attraction alone does not prove a magnet.
5	**(c)** plants need light for healthy growth	Without light, plants turn pale.
6	**(a)** crocodile	Crocodiles live both in water and land.
7	**(d)** vegetables and fruits	They are protective foods.
8	**(a)** iron, nickel, cobalt	Standard magnetic materials.
9	**(a)** The number alone has no meaning without it	Units give meaning.
10	**(d)** herb, shrub, tree	Herbs are smallest.
11	**(c)** list of ingredients	Ingredients tell what we eat.
12	**(c)** push away from each other	Like poles repel.
13	**(a)** the presence of water only	Only one variable changes at a time.
14	**(a)** herb	Soft stems and low height.
15	**(b)** roti, dal, vegetables and curd	Variety of nutrients.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Then the effect can be linked to that one cause (1); It makes results clear and fair (1).	1+1
17	To reduce loss of water (1); The green stem makes the food (1).	1+1
18	Proteins and vitamins (1); Minerals and fibre are also low (1).	1+1
19	No, a magnet repels one pole (1); The rod is a magnetic material only (1).	1+1
20	It collects light from far objects (1); It makes faraway objects look bigger and brighter (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Take two sets of identical seeds with same moisture (1); Keep one set in a warm and another in a cool place (1); Compare sprouting after fixed days (1).	1+1+1
22	Lotus: water (1); Cactus: desert (1); Pine: mountain (1).	1+1+1
23	It saves time but lacks nutrients (1); It has too much fat, sugar and salt (1); Regular use harms health (1).	1+1+1
24	Use a magnet to remove iron (1); Add water to dissolve salt, filter sand (1); Evaporate water to get salt (1).	1+1+1
	OR: Use same size and number of sheets (1); Pour same amount of water on each (1); Compare water absorbed (1).	1+1+1
25	Gills (1); Fins and tail (1); Streamlined body (1).	1+1+1
	OR: Dehydration (1); Wastes not removed (1); Tiredness and headache (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Change: length of the string (1); Keep same: bob, angle of release (1); Measure time for 10 swings (1); Repeat and average (1); Record in a table and conclude (1).	1+1+1+1+1
	OR: Camel: hump and broad feet (1); Fish: gills and fins (1); Bird: wings and light bones (1); Polar bear: thick fur and fat (1); Each helps survival in its habitat (1).	1+1+1+1+1

27	Reduce junk food (1); Add fruits and vegetables (1); Add dal or eggs for protein (1); Add milk for calcium (1); Encourage water and exercise (1).	1+1+1+1+1
	OR: Hang one bar magnet (1); Bring N of other near N (1); Observe repulsion (1); Bring S near N: attraction (1); Conclude: like repel, unlike attract (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** north pole	Like poles repel.
2	**(a)** placing the ruler's zero mark at one end and reading straight on	Correct use of the scale.
3	**(b)** to reduce heat loss	Insulation.
4	**(b)** spinach	It provides iron and vitamins.
5	**(c)** aluminium	Aluminium is not attracted.
6	**(a)** keeping the eye directly above the mark	Avoid parallax by looking straight.
7	**(a)** classification	It helps to organise.
8	**(d)** expiry date	Eating expired food is unsafe.
9	**(d)** unlike poles	Unlike attract.
10	**(c)** amount of water and sunlight	Other conditions are kept constant.
11	**(a)** shrub	Characteristic of a shrub.
12	**(a)** vitamins and minerals	Vitamins and minerals regulate body functions.
13	**(b)** unlike	Opposite poles attract.
14	**(c)** listen to heartbeats	A stethoscope amplifies body sounds.
15	**(b)** thick stem and spiny leaves	Reduces water loss.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	A bat is a mammal, which gives birth and feeds milk (1); A bat has skin wings rather than feathers (1).	1+1
17	Difficulty seeing in dim light (1); It may lead to night blindness (1).	1+1
18	They repel (1); Like poles repel (1).	1+1
19	Prediction: what we expect to happen (1); Observation: what actually happens (1).	1+1
20	Hold the soil firmly (1); Prevent soil erosion (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Bring the marked N pole near an end (1); If attracted, that end is S (1); If repelled, that end is N (1).	1+1+1
22	Chemicals or flame may cause harm (1); Safety rules must be known first (1); The teacher's guidance prevents accidents (1).	1+1+1
23	Help in flight (1); Keep the body warm (1); Streamline the body (1).	1+1+1
24	Carrot (1); Papaya or mango (1); Green leafy vegetables (1).	1+1+1
	OR: Aluminium: foils (1); Copper: wires (1); Gold: jewellery (1).	1+1+1
25	Popularity is not evidence (1); Claims must be tested (1); Scientists accept only what experiments support (1).	1+1+1
	OR: Mango has tap root (1); Reticulate leaf venation (1); It is a tree (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Add the readings: 480 cm (1); Average = 480 / 4 = 120 cm (1); Repeating reduces random error (1); Readings vary a little (1); Average is more reliable (1).	1+1+1+1+1
	OR: Groups give protection, e.g. deer (1); Group hunting, e.g. wolves (1); Some animals live alone, e.g. tiger (1); Need large territory (1); Depends on food and safety (1).	1+1+1+1+1

27	Heat destroys vitamin C (1); Boiling vegetables loses water-soluble vitamins (1); Steaming retains nutrients (1); Cooking makes some foods digestible (1); Do not overcook (1).	1+1+1+1+1
	OR: Both attracted by the bar (1); Magnet may be repelled (1); Iron is not repelled (1); Test confirms magnet (1); Iron may become magnetised (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** to retain vitamins	Heat destroys vitamins.
2	**(d)** only some metals are magnetic	Only some are magnetic.
3	**(d)** two things changed at once, so the cause is unclear	Only one variable should change.
4	**(b)** the name given by a person	Names are labels not features.
5	**(b)** we may eat too much without noticing	Attention is on the screen.
6	**(a)** in a box with a keeper	Keepers prevent loss.
7	**(d)** observation, hypothesis, experiment, conclusion	Standard order of the scientific method.
8	**(c)** fibrous roots	Parallel venation goes with fibrous roots.
9	**(d)** iron	Anaemia gives paleness and tiredness.
10	**(a)** north-seeking pole	North pole.
11	**(b)** measure body temperature	It records temperature.
12	**(d)** small or reduced	They absorb water directly.
13	**(b)** whole fruits have fibre	Fibre is retained.
14	**(c)** a magnet can attract iron without touching it	Force acts from a distance.
15	**(d)** The ice cube will melt faster on a metal plate.	A prediction is about a future result.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	They give different vitamins and minerals (1); They give variety (1).	1+1
17	Pole (1); Magnetic force is concentrated there (1).	1+1
18	No, it is based on appearance only (1); More evidence from observations shows the Earth moves (1).	1+1
19	Camouflage in snow helps hunting (1); Thick fur keeps them warm (1).	1+1
20	It increases vitamins (1); It makes digestion easier (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Roti or rice for energy (1); Dal or paneer for protein (1); Vegetable and fruit for vitamins (1).	1+1+1
22	Magnetise the needle (1); Float it on a cork in water (1); It settles in north-south (1).	1+1+1
23	It may have rained (1); A water tanker may have spilt (1); More evidence is required (1).	1+1+1
24	Little water (1); Very hot days (1); Poor soil (1).	1+1+1
	OR: Milk and curd (1); Sunlight for vitamin D (1); Green leafy vegetables (1).	1+1+1
25	It is pushed by Earth's field (1); Overshoots north-south (1); Friction stops it finally (1).	1+1+1
	OR: Extend senses (1); Give accurate values (1); Allow study of very small or very far objects (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Form from observation: plants in shade look pale (1); Hypothesis: plants need sunlight (1); Test with plants in light and dark (1); Pale plants in dark support it (1); Repeat results, then accept (1).	1+1+1+1+1
	OR: Mango: net venation (1); Mango: tap root (1); Grass: parallel venation (1); Grass: fibrous roots (1); Reason: different groups of plants (1).	1+1+1+1+1

27	Starch: iodine on potato blue-black (1); Fat: paper test with butter (1); Protein: copper sulphate and sodium hydroxide with egg white (1); Observe each (1); Record in a table (1).	1+1+1+1+1
	OR: (a) Force passes through paper (1); (b) Copper is non-magnetic (1); (c) Pairs with keepers preserve magnetism (1); Pole arrangement unlike (1); Prevents weakening (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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